

PC-03 · EMULSIFIERS & THICKENERS · PERSONAL CARE

# HS-C500

Carbomer

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| INCI<br>Carbomer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CAS<br>9007-20-9                                                                          | EC<br>— | FAMILY<br>PC-03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SUPPLIED AS<br>Powder |           |        |                 |                                                                                           |            |                     |                               |                                           |                |      |                  |                   |
| <div>01 POSITIONING</div> <p>Four grades, one rheology language — pick the curve, we ship the powder that draws it.</p> <div>02 HOW IT WORKS</div> <div>01 Swell on neutralization</div> <p>Neutralized carboxyls repel, swelling each microgel up to 1000× — viscosity from 0.2% dosage.</p> <div>02 Yield stress</div> <p>Suspends beads, pigments and bubbles indefinitely, yet flows instantly under shear.</p> <div>03 Crystal clarity</div> <p>Fully hydrated gels transmit light — the transparent-gel category exists on this polymer.</p> <div>03 APPLICATIONS</div> <div>→ Hydroalcoholic and moisturizing gels</div> <div>→ Bead-suspension shower gels</div> <div>→ Styling gels</div> <div>→ Toothpastes (grade-dependent)</div> <div>→ Emulsion stabilization</div> |                                                                                           |         | <div>04 FORMULATION GUIDE</div> <table><tr><td>USE LEVEL</td><td>0.2–1%</td></tr><tr><td>PH &amp; CONDITIONS</td><td>Neutralize to pH 5.5–7 (NaOH, TEA, arginine); viscosity collapses below pH 4 and above 10</td></tr></table> <p>— Disperse slowly into vortexing water before neutralizing; hydrate fully to avoid fish-eyes.</p> <p>— Electrolytes thin carbomer gels — for salt-rich systems choose 702/500-type grades.</p> <div>05 TYPICAL SPECIFICATION</div> <table><tr><td>Appearance</td><td>White fluffy powder</td></tr><tr><td>Viscosity (0.5%, neutralized)</td><td>per grade: 40 000–60 000 mPa·s (940-type)</td></tr><tr><td>Loss on drying</td><td>≤ 2%</td></tr><tr><td>Residual benzene</td><td>≤ 2 ppm (per TDS)</td></tr></table> <p>Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.</p> <div>06 PERFORMANCE — DIRECTIONAL</div> <p>Viscosity per 0.5% dosage<br/>Carbomer 940-type</p> <div></div> <p>Suspension of 1 mm beads (30 d)<br/>Carbomer gel</p> <div></div> <p>Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.</p> |                       | USE LEVEL | 0.2–1% | PH & CONDITIONS | Neutralize to pH 5.5–7 (NaOH, TEA, arginine); viscosity collapses below pH 4 and above 10 | Appearance | White fluffy powder | Viscosity (0.5%, neutralized) | per grade: 40 000–60 000 mPa·s (940-type) | Loss on drying | ≤ 2% | Residual benzene | ≤ 2 ppm (per TDS) |
| USE LEVEL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.2–1%                                                                                    |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |           |        |                 |                                                                                           |            |                     |                               |                                           |                |      |                  |                   |
| PH & CONDITIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Neutralize to pH 5.5–7 (NaOH, TEA, arginine); viscosity collapses below pH 4 and above 10 |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |           |        |                 |                                                                                           |            |                     |                               |                                           |                |      |                  |                   |
| Appearance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | White fluffy powder                                                                       |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |           |        |                 |                                                                                           |            |                     |                               |                                           |                |      |                  |                   |
| Viscosity (0.5%, neutralized)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | per grade: 40 000–60 000 mPa·s (940-type)                                                 |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |           |        |                 |                                                                                           |            |                     |                               |                                           |                |      |                  |                   |
| Loss on drying                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ≤ 2%                                                                                      |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |           |        |                 |                                                                                           |            |                     |                               |                                           |                |      |                  |                   |
| Residual benzene                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ≤ 2 ppm (per TDS)                                                                         |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |           |        |                 |                                                                                           |            |                     |                               |                                           |                |      |                  |                   |